MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 1

1. *Consider the following two linear equations:*

(a) $$2x_1 + 5x_2 = 2$$

(b) $$2x_1 + 5x_2 = 0.$$

If (x_1, x_2) and (x_1', x_2') are solutions to (a), is $(x_1 + x_1', x_2 + x_2')$ also a solution to (a)? If (x_1, x_2) and (x_1', x_2') are solutions to (b), is $(x_1 + x_1', x_2 + x_2')$ also a solution to (b)?

2. *Consider the following system:*

(c) $$\begin{aligned} a_{1,1}x_1 + \cdots + a_{1,n}x_n &= b_1 \\ \vdots \qquad\qquad \vdots \quad &= \vdots \\ a_{m,1}x_1 + \cdots + a_{m,n}x_n &= b_m \end{aligned}$$

Answer the following questions!

(1) *If $(x_1, \ldots, x_n)$ and $(x_1', \ldots, x_n')$ are solutions to (c), is $(x_1 + x_1', \ldots, x_n + x_n')$ also a solution to (c)?*

(2) *What conditions on $b_1, \ldots, b_m$ can you impose such that (1) becomes true?*

3. *What is the largest possible number of pivots a 4×6 matrix can have?*

4. *What is the largest possible number of pivots a 6×4 matrix can have?*

5. *How many solutions does a consistent linear system of 3 equations and 4 unknowns have?*

6. *Just using elementary row operations, can you transform* $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ *into* $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$*?*

7. *Let $A = [a_{ij}]_{3\times 4}$, and let $B = [b_{ij}]_{3\times 4}$ be an echelon form of A.*

7.1. *Is it true that, if $a_{11} = 0$, then $b_{11} = 0$?*

7.2. *Is it true that, if A has a column of zeros, then B also has a column of zeros?*

7.3. *Suppose B has a row of zeros. What can you say about rows of A?*

8. *Describe all 2×2-matrices that are of reduced echelon form!*

9. *True or false? If true, explain why! If false, give a counterexample.*

(1) *There is no more than one pivot position in any row.*

(2) *There is no more than one pivot position in any column.*

(3) *There cannot be more free variables than pivot variables.*

(4) *There is a linear system that has exactly two solutions.*

(5) *If the augmented matrix of a linear system has two identical rows, the linear system is necessarily inconsistent.*

Solutions will be posted at the end of Week 2.